

Replication Project: Oil, Islam, and Women

Kate Cox

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Original Study

Theory

The study *Oil, Islam, and Women* by Michael L. Ross proposes that women in the Middle East have been held back from the workforce not by Islamic traditions, as has been the primary theory in the literature, but by oil. This questions another idea documented in existing literature, that economic growth benefits gender equality.

The study proposes that different types of economic growth have different impacts on gender equality in the labor force. He proposes that oil and mineral extraction leaves few opportunities for women, leading to impacts on their labor and, therefore, social and political inclusion. He believes that, in contrast to the negative effect of an oil-based economy on women, export-oriented industries, specifically textiles, are positive for women's economic growth. He suggests that, if these findings hold, they may have implications for the way that the U.S. considers oil transactions.

Method

Ross analyzed oil production and female employment data from 1960 until 2002. He looked at first-differences models with fixed effects to examine variations over time in the same state. He also looked at cross-national models to examine variations between states. His tables include both GLS and OLS regressions.

Variables

The outcome variable is *Oil Rents Per Capita*. The predictor variables are *Female Labor Force Participation* and female political influence measured by *Female Seats* (the percentage of seats held by women in the country's parliament) and *Female Ministers* (the percentage of the ministerial positions held by women). This study uses 2002 data. The study controls for other variables including *Income*, *Income Squared*, *Middle East* (whether the country is

in the Middle East), *Islam* (the proportion of the country that is Muslim), *Communist* (a dummy variable indicating whether or not the country is Communist), *Working Age* (the proportion of the population between 15 and 64), *Proportional Representation* (a dummy variable for states with proportional representation in parliament), *Closed List*, (a dummy variable for electoral systems with closed lists), and *Polity* (a scale to measure democracy level).

Data

There are 2 data sources provided for replication, `data_long` and `data_short`.

```
1 data_long <- read.dta("Replication data - long.dta")
2 data_short <- data.frame(read.dta("Replication data - short.dta"))
```

Table 1

```
1 # Prepare data for GLS to match the original study
2 data_gls <- data_long %>%
3   filter(!is.na(zFDlaborfemale), !is.na(zFDlogGDPcap_1)) %>%
4   group_by(id2) %>%
5   arrange(year) %>%
6   slice(-1) %>%
7   ungroup() %>%
8   mutate(id2.factor = as.factor(id2))
9
10 # Model 1.1
11 model_1.1 <- gls(zFDlaborfemale ~ zFDlogGDPcap_1 + zFDlogGDPcapSQ_1 + zFDage_1
12   + id2.factor,
13   data = data_gls,
14   correlation = corAR1(0.1, form = ~year|id2.factor), method =
15   "ML")
16
17 # Model 1.2
18 model_1.2 <- update(model_1.1, . ~ . + zFDoil_gas_rentPOP_1)
19
20 # Exclude Kuwait and Saudi Arabia
21 data_gls_no_oil <- data_gls %>% filter(!id %in% c("KWT", "SAU"))
22
23 # Model 1.3
24 model_1.3 <- gls(zFDlaborfemale ~ zFDlogGDPcap_1 + zFDlogGDPcapSQ_1 + zFDage_1
25   +
26   zFDoil_gas_rentPOP_1 + id2.factor,
27   data = data_gls_no_oil,
28   correlation = corCAR1(0.2, form = ~year|id2.factor), method =
29   "ML")
30
31 # Prepare data for OLS to match the original study
32 data_ols <- data_long %>%
33   filter(!is.na(zFDlaborfemale), !is.na(zFDlogGDPcap_1)) %>%
34   mutate(id2.factor = as.factor(id2), year.factor = as.factor(year))
```

```

31
32 # Model 1.4
33 model_1.4 <- lm(zFDlaborfemale ~ zFDlogGDPcap_1 + zFDlogGDPcapSQ_1 + zFDage_1
34               +
35               zFDoil_gas_rentPOP_1 + id2.factor + year.factor, data = data
36               _ols)
37
38 #Table 1 Stargazer
39 stargazer(model_1.1, model_1.2, model_1.3, model_1.4,
40           type = "latex",
41           title = "TABLE 1. Pooled Time-Series Cross-national Regressions,
42           with First Differences and Fixed Effects",
43           dep.var.labels = "Female Labor Force Participation, 1960–2002",
44           covariate.labels = c("Income (log)",
45                               "Income squared (log)",
46                               "Working Age",
47                               "Oil Rents per capita"),
48           omit = c("id2.factor", "year.factor"),
49           omit.stat = c("f", "ser"))

```

Table 2

```

1 # Models 2.1 – 2.4
2 model_2.1 <- lm(zlaborfemaleMOD ~ zlogGDPcap + zlogGDPcap2 + zage15_64 + zme_
3   nafr + zcommie, data = data_short)
4 model_2.2 <- update(model_2.1, . ~ . + zislam)
5 model_2.3 <- update(model_2.1, . ~ . + zoil_gas_rentPOP)
6 model_2.4 <- update(model_2.1, . ~ . + zislam + zoil_gas_rentPOP)
7
8 # Robust SEs for Table 2
9 model_2_list <- list(model_2.1, model_2.2, model_2.3, model_2.4)
10 model_2_se <- lapply(model_2_list, function(x) sqrt(diag(vcovHC(x, type = "HCl
11   "))))
12
13 stargazer(model_2_list, se = model_2_se,
14           type="latex",
15           column.sep.width = "0",
16           font.size = "small",
17           df = FALSE,
18           title="Table 2. Cross-national Regressions on Female Labor Force",
19           dep.var.labels = "Female Nonagricultural Labor Force Participation,
20           1993–2002",
21           covariate.labels = c("Income (log)",
22                               "Income squared (log)",
23                               "Working Age",
24                               "Middle East",
25                               "Communist",
26                               "Islam",
27                               "Oil Rents per capita"))

```

Table 1. Pooled Time-Series Cross-national Regressions, with First Differences and Fixed Effects

	<i>Dependent variable:</i>			
	Female Labor Force Participation, 1960-2002			
		<i>generalized least squares</i>		<i>OLS</i>
	(1)	(2)	(3)	(4)
Income (log)	−0.014 (0.033)	−0.042 (0.034)	−0.013 (0.028)	−0.051 (0.047)
Income squared (log)	0.018 (0.033)	0.051 (0.034)	0.018 (0.029)	0.021 (0.048)
Working Age	0.107*** (0.022)	0.106*** (0.022)	0.067*** (0.021)	0.177*** (0.013)
Oil Rents per capita		−0.028*** (0.007)	−0.017* (0.010)	−0.049*** (0.011)
Constant	−0.498 (0.364)	−0.497 (0.364)	−0.536 (0.400)	−0.404** (0.179)
Observations	5,234	5,234	5,168	5,395
R ²				0.562
Adjusted R ²				0.544
Log Likelihood	−3,536.648	−3,528.950	−2,474.278	
Akaike Inf. Crit.	7,405.295	7,391.900	5,278.555	
Bayesian Inf. Crit.	8,494.742	8,487.910	6,359.345	

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 2. Cross-national Regressions on Female Labor Force

	<i>Dependent variable:</i>			
	Female Nonagricultural Labor Force Participation, 1993-2002			
	(1)	(2)	(3)	(4)
Income (log)	-1.334 (0.843)	-1.499* (0.874)	-1.757** (0.852)	-1.864** (0.878)
Income squared (log)	1.615** (0.798)	1.735** (0.823)	2.053** (0.805)	2.122** (0.824)
Working Age	-0.386*** (0.141)	-0.395*** (0.143)	-0.340** (0.139)	-0.350** (0.142)
Middle East	-0.521*** (0.083)	-0.407*** (0.117)	-0.413*** (0.084)	-0.326*** (0.117)
Communist	0.290*** (0.104)	0.292*** (0.104)	0.284*** (0.103)	0.286*** (0.104)
Islam		-0.170 (0.113)		-0.139 (0.116)
Oil Rents per capita			-0.225*** (0.053)	-0.210*** (0.055)
Constant	-0.012 (0.062)	-0.014 (0.062)	-0.010 (0.060)	-0.012 (0.060)
Observations	167	167	167	167
R ²	0.382	0.396	0.417	0.427
Adjusted R ²	0.363	0.374	0.395	0.401
Residual Std. Error	0.800	0.793	0.780	0.776
F Statistic	19.889***	17.501***	19.086***	16.898***

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 3

```

1 # Middle Eastern, Islamic, and Developed nation vectors
2 islamic <- c("ARE", "BHR", "OMN", "QAT", "DZA", "EGY", "IRN", "IRQ", "JOR", "
  LBN", "MAR", "SYR", "TUN", "YEM", "AFG", "AZE", "BFA", "BGD", "BRN", "COM",
  "DJI", "GIN", "GMB", "IDN", "KGZ", "MDV", "MLI", "MRT", "MYS", "NER", "
  PAK", "SDN", "SEN", "SLE", "SOM", "TCD", "TJK", "TKM", "TUR", "UZB")
3 developed <- c("USA", "CAN", "JPN", "GBR", "FRA", "GER", "ITA", "ESP", "SWE",
  "NOR", "DNK", "NLD", "BEL", "AUT", "CHE", "FIN", "IRL", "PRT", "GRC", "AUS",
  "NZZ", "ISL", "LUX")
4
5 data_table_3 <- data_short %>%
6   filter(!id %in% c("SAU", "KWT"), !is.na(female_seats)) %>%
7   mutate(
8     oil_rich = ifelse(zoil_gas_rentPOP >= quantile(zoil_gas_rentPOP, 1-(38/
9       159), na.rm=T), "Oil-rich", "Oil-poor"),
10    income_status = ifelse(zlogGDPcap > -0.12, "High Income", "Low Income"),
11    is_me = (me_nafr == 1), is_islamic = id %in% islamic, is_dev = !id %in%
12    developed
13  )
14 get_rep_mean <- function(df, row_filt, col_val) {
15   val <- df %>% filter(!enquo(row_filt), oil_rich == col_val) %>% summarise(m
16     = mean(female_seats, na.rm = TRUE)) %>% pull(m)
17   return(round(val, 1))
18 }
19 table3_calculated <- data.frame(
20   Group = c("High Income", "Low Income", "Middle East", "Non-Middle East (all)",
21     "Non-Middle East (dev)", "Islamic", "Non-Islamic (all)", "Non-Islamic (
22     dev)"),
23   Oil_Rich = c(get_rep_mean(data_table_3, income_status="High Income", "Oil-
24     rich"), get_rep_mean(data_table_3, income_status="Low Income", "Oil-rich"
25     ), get_rep_mean(data_table_3, is_me, "Oil-rich"), get_rep_mean(data_table_
26     3, !is_me, "Oil-rich"), get_rep_mean(data_table_3 %>% filter(is_dev), !is_
27     me, "Oil-rich"), get_rep_mean(data_table_3, is_islamic, "Oil-rich"), get_
28     rep_mean(data_table_3, !is_islamic, "Oil-rich"), get_rep_mean(data_table_3
29     %>% filter(is_dev), !is_islamic, "Oil-rich")),
30   Oil_Poor = c(get_rep_mean(data_table_3, income_status="High Income", "Oil-
31     poor"), get_rep_mean(data_table_3, income_status="Low Income", "Oil-poor"
32     ), get_rep_mean(data_table_3, is_me, "Oil-poor"), get_rep_mean(data_table_
33     3, !is_me, "Oil-poor"), get_rep_mean(data_table_3 %>% filter(is_dev), !is_
34     me, "Oil-poor"), get_rep_mean(data_table_3, is_islamic, "Oil-poor"), get_
35     rep_mean(data_table_3, !is_islamic, "Oil-poor"), get_rep_mean(data_table_3
36     %>% filter(is_dev), !is_islamic, "Oil-poor"))
37 )
38 print("TABLE 3. Parliamentary Seats Held by Woman, 2002 (percentage)"); print(
39   table3_calculated)

```

Note: Attempting to replicate Table 3 (which was not included in the Anderson and

Hill replication code) revealed problems with Ross's provided replication data and lack of transparency. For many variables, he only provided the Z-scores, not the raw numbers. This made it impossible to replicate his original method as he did not clarify which countries he put in each category. This code strikes a balance between constraining itself so that it is performing similarly to Ross's (making similar choices about what countries to drop and which countries belong in each category), while still performing actual calculations using the data available.

[1] "TABLE 3. Parliamentary Seats Held by Woman, 2002 (percentage)"

Group	Oil_Rich	Oil_Poor		
1	High Income		15.7	17.9
2	Low Income		11.6	13.0
3	Middle East		5.5	6.5
4	Non-Middle East (all)		17.8	15.7
5	Non-Middle East (dev)		13.9	14.5
6	Islamic		7.2	9.8
7	Non-Islamic (all)		19.3	16.5
8	Non-Islamic (dev)		15.0	15.2

Table 4

```

1 model_4.1 <- lm(zfemale_seats ~ zlogGDPcap + zme_nafr, data = data_short)
2 model_4.2 <- update(model_4.1, . ~ . + zislam)
3 model_4.3 <- update(model_4.1, . ~ . + zoil_gas_rentPOP)
4 model_4.4 <- update(model_4.1, . ~ . + zislam + zoil_gas_rentPOP)
5 model_4.5 <- update(model_4.4, . ~ . + zpolity)
6 model_4.6 <- update(model_4.5, . ~ . + zPR_MOD)
7 model_4.7 <- update(model_4.6, . ~ . + zclosed_list)
8 model_4.8 <- update(model_4.4, . ~ . + zlaborfemaleMOD)
9
10 model_4_list <- list(model_4.1, model_4.2, model_4.3, model_4.4, model_4.5,
11                      model_4.6, model_4.7, model_4.8)
12 model_4_se <- lapply(model_4_list, function(x) sqrt(diag(vcovHC(x, type = "HC1"
13                      )))))
14
15 stargazer(model_4_list,
16           se = model_4_se,
17           type="latex",
18           title="Table 4. Cross-national Regressions on Female Seats in
19           Parliament",
20           font.size = "small",
21           column.sep.width = "-15pt",
22           df = FALSE,
23           dep.var.labels = "Parliamentary Seats Held by Women (percent), 2002",
24           ,
25           covariate.labels = c("Income (log)",
26                                "Middle East",
27                                "Islam",

```

```

24         "Oil Rents per Capita",
25         "Polity",
26         "Proportional Representation",
27         "Closed List",
28         "Female Labor Force Participation"))

```

See Table 4 displayed below, on page 9.

Table 5

```

1  model_5.1 <- lm(zfemale_min ~ zlogGDPcap + zme_nafr, data = data_short)
2  model_5.2 <- update(model_5.1, . ~ . + zislam)
3  model_5.3 <- update(model_5.1, . ~ . + zoil_gas_rentPOP)
4  model_5.4 <- update(model_5.1, . ~ . + zislam + zoil_gas_rentPOP)
5  model_5.5 <- update(model_5.4, . ~ . + zlaborfemaleMOD)
6
7  model_5_list <- list(model_5.1, model_5.2, model_5.3, model_5.4, model_5.5)
8  model_5_se <- lapply(model_5_list, function(x) sqrt(diag(vcovHC(x, type = "HC1"
9  "))))
10 stargazer(model_5_list,
11           se = model_5_se,
12           type="latex",
13           df = FALSE,
14           title="Table 5. Cross-national Regressions on Female Ministerial
15           Positions 2002",
16           dep.var.labels = "Female Ministerial Positions 2002",
17           covariate.labels = c("Income (log)",
18                               "Middle East",
19                               "Islam",
20                               "Oil Rents per Capita",
21                               "Female Labor Force Participation"))

```


Table 4. Cross-national Regressions on Female Seats in Parliament

	<i>Dependent variable:</i>							
	Parliamentary Seats Held by Women (percent), 2002							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Income (log)	0.311*** (0.082)	0.259*** (0.092)	0.365*** (0.083)	0.320*** (0.094)	0.390*** (0.094)	0.367*** (0.089)	0.779*** (0.137)	0.329*** (0.088)
Middle East	-0.378*** (0.062)	-0.261*** (0.084)	-0.278*** (0.060)	-0.193** (0.080)	-0.233*** (0.083)	-0.253*** (0.077)	-0.182 (0.143)	-0.080 (0.095)
Islam		-0.178* (0.092)		-0.139 (0.089)	-0.180* (0.097)	-0.144 (0.096)	-0.123 (0.175)	-0.103 (0.087)
Oil Rents per Capita			-0.233*** (0.070)	-0.218*** (0.066)	-0.258*** (0.076)	-0.238*** (0.069)	-0.317*** (0.087)	-0.152** (0.070)
Polity					-0.158 (0.103)	-0.298*** (0.101)	-0.294** (0.149)	
Proportional Representation						0.316*** (0.073)	0.013 (0.133)	
Closed List							0.268*** (0.091)	
Female Labor Force Participation								0.309*** (0.084)
Constant	-0.013 (0.070)	-0.018 (0.069)	-0.015 (0.068)	-0.018 (0.068)	-0.016 (0.068)	-0.022 (0.064)	0.034 (0.105)	-0.015 (0.066)
Observations	161	161	161	161	161	161	88	160
R ²	0.212	0.227	0.255	0.264	0.275	0.346	0.403	0.328
Adjusted R ²	0.202	0.212	0.240	0.245	0.252	0.320	0.350	0.306
Residual Std. Error	0.893	0.888	0.872	0.869	0.865	0.824	0.828	0.835
F Statistic	21.271***	15.374***	17.885***	13.966***	11.754***	13.562***	7.702***	15.027***

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5. Cross-national Regressions on Female Ministerial Positions 2002

	<i>Dependent variable:</i>				
	Female Ministerial Positions 2002				
	(1)	(2)	(3)	(4)	(5)
Income (log)	0.250*** (0.089)	0.239** (0.096)	0.282*** (0.094)	0.278*** (0.102)	0.281*** (0.102)
Middle East	-0.244*** (0.049)	-0.218*** (0.068)	-0.181*** (0.038)	-0.174*** (0.058)	-0.141** (0.061)
Islam		-0.039 (0.081)		-0.012 (0.081)	-0.002 (0.082)
Oil Rents per Capita			-0.139** (0.061)	-0.138** (0.062)	-0.117* (0.064)
Female Labor Force Participation					0.094 (0.075)
Constant	-0.012 (0.075)	-0.014 (0.074)	-0.012 (0.075)	-0.013 (0.074)	-0.012 (0.074)
Observations	155	155	155	155	154
R ²	0.105	0.106	0.121	0.121	0.125
Adjusted R ²	0.094	0.088	0.103	0.098	0.095
Residual Std. Error	0.952	0.955	0.947	0.950	0.953
F Statistic	8.949***	5.969***	6.922***	5.160***	4.222***

Note: *p<0.1; **p<0.05; ***p<0.01

Table 6

```

1 econ_2003 <- data_long %>%
2   filter(id %in% c("DZA", "MAR", "TUN"), year == 2003) %>%
3   mutate(Income = round(exp(logGDPcap), 0), Oil = round(oil_gas_rentPOP, 0))
4   %>%
5   select(id, Income, Oil)
6
7 table_6_calc <- data_short %>%
8   filter(id %in% c("DZA", "MAR", "TUN")) %>%
9   left_join(econ_2003, by = "id") %>%
10  mutate(
11    Country = case_when(id == "DZA" ~ "Algeria", id == "MAR" ~ "Morocco", id
12      == "TUN" ~ "Tunisia"),
13    'Muslim population (%)' = case_when(id == "DZA" ~ 97, id == "MAR" ~ 98, id ==
14      "TUN" ~ 99),
15    'Fertility Rate' = case_when(id == "DZA" ~ 2.72, id == "MAR" ~ 2.66, id ==
16      "TUN" ~ 1.99),
17    'Textile exports per capita' = case_when(id == "DZA" ~ 0.09, id == "MAR" ~
18      94, id == "TUN" ~ 287),
19    'Income per capita' = Income, 'Oil rents per capita' = Oil,
20    'Female Labor (%)' = round(laborfemaleMOD, 1), 'Female Seats (%)' = round(
21      female_seats, 1),
22    'Gender Rights Index' = round(gender_rights, 2)
23  ) %>%
24  select(Country, 'Muslim population (%)', 'Income per capita', 'Oil rents per
25    capita', 'Textile exports per capita', 'Female Labor (%)', 'Female Seats
26    (%)', 'Fertility Rate', 'Gender Rights Index')
27
28 table_6_final <- table_6_calc %>%
29   mutate(across(everything(), as.character)) %>%
30   pivot_longer(cols = -Country, names_to = "Variable") %>%
31   pivot_wider(names_from = Country) %>%
32   mutate(Algeria = ifelse(Variable == "Female Labor (%)", "12 (est.)", Algeria
33     ), Tunisia = ifelse(Variable == "Female Labor (%)", "n.a.", Tunisia))
34
35 print("Table 6. Comparison of Algeria, Morocco, and Tunisia")
36 print(table_6_final)

```

Variable	Algeria	Morocco	Tunisia
<chr>	<chr>	<chr>	<chr>
1 Muslim population (%)	97	98	99
2 Income per capita	1974	1339	2225
3 Oil rents per capita	863	0	52
4 Textile exports per capita	0.09	94	287
5 Female Labor (%)	12 (est.)	32.8	n.a.
6 Female Seats (%)	6.2	10.8	22.8
7 Fertility Rate	2.72	2.66	1.99
8 Gender Rights Index	2.82	3.1	3.44
8 Gender Rights Index	2.82	3.1	3.44

As with table 3, exact proportions were not possible without the raw data. However, this table gave numbers and findings very similar to the ones in Ross's original study.

Figure 2

Note: Figure 1 is excluded because it is a flow chart, not a plot of the data

```
1 raw_oil_2002 <- data_long %>% filter(year == 2002) %>% select(id, oil_gas_rentPOP)
2 plot_data_f2 <- data_short %>% left_join(raw_oil_2002, by = "id") %>% filter(!
  id %in% c("SAU", "KWT"), !is.na(oil_gas_rentPOP), !is.na(female_seats))
3
4 jpeg(file="figure2.jpeg")
5 ggplot(plot_data_f2, aes(x = oil_gas_rentPOP, y = female_seats)) +
6   geom_smooth(method = "lm") +
7   coord_cartesian(xlim = c(0, 6000), ylim = c(0, 20)) +
8   labs(title = "Effect of Oil Rents on Female Parliamentary Seats", x = "Oil
  and Gas Rents per Capita (USD)", y = "Women in Parliament (%)")
9 dev.off()
```

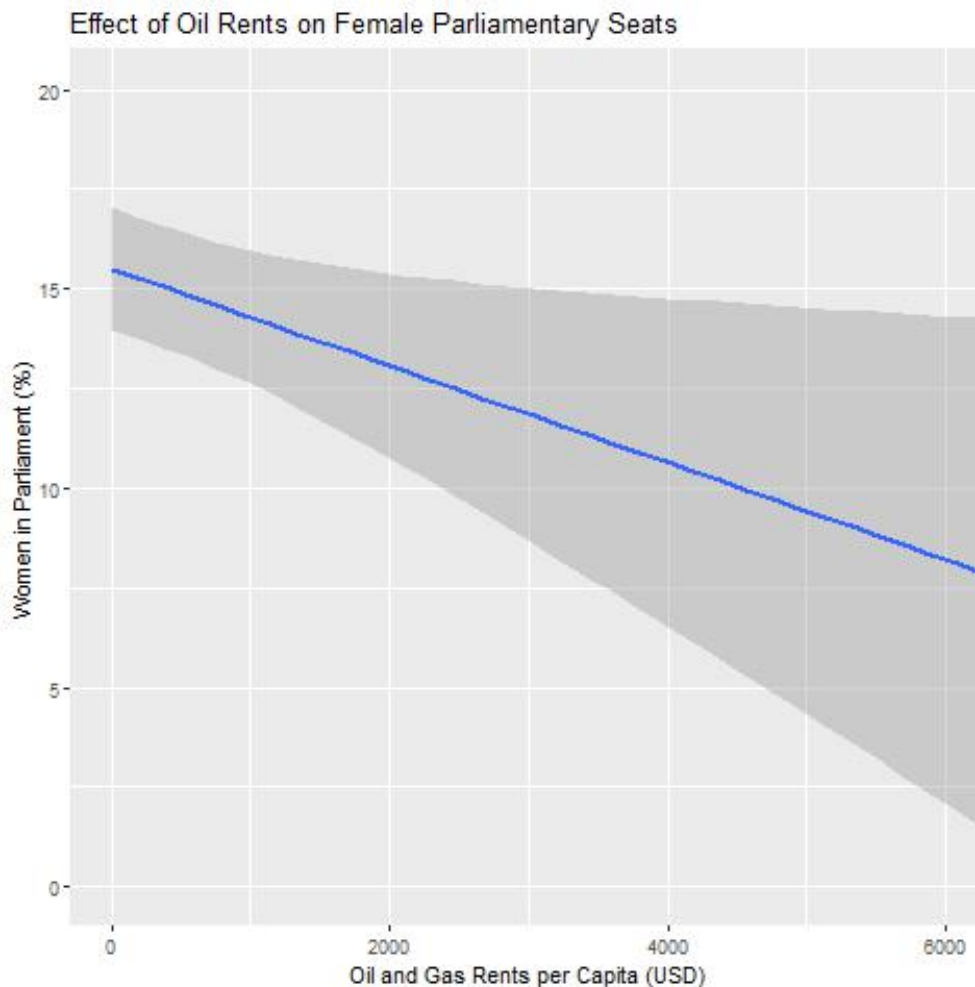


Figure 3

```

1 oil_averages <- data_long %>% filter(year >= 1993, year <= 2002) %>% group_by(
  id) %>% summarise(raw_oil = mean(oil_gas_rentPOP, na.rm = TRUE))
2
3 jpeg(file="figure3.jpeg")
4 data_short %>% filter(me_nafr == 1) %>% left_join(oil_averages, by = "id") %>%
5 ggplot(aes(x = raw_oil, y = laborfemaleMOD, label = id)) +
6 geom_point(color = "black", size = 2) + geom_text(size = 2.5, vjust = -1,
  hjust = 0.5) +
7 scale_x_continuous(name = "Oil and Gas Rents per capita, 1993-2002", breaks
  = seq(0, 15000, 5000)) +
8 scale_y_continuous(name = "Female Percent of Nonagricultural Labor Force,
  1993-2002", breaks = seq(0, 40, 10)) +
9 coord_cartesian(xlim = c(0, 15000), ylim = c(0, 40)) + theme_bw() + theme(
  panel.grid.minor = element_blank())
10 dev.off()

```

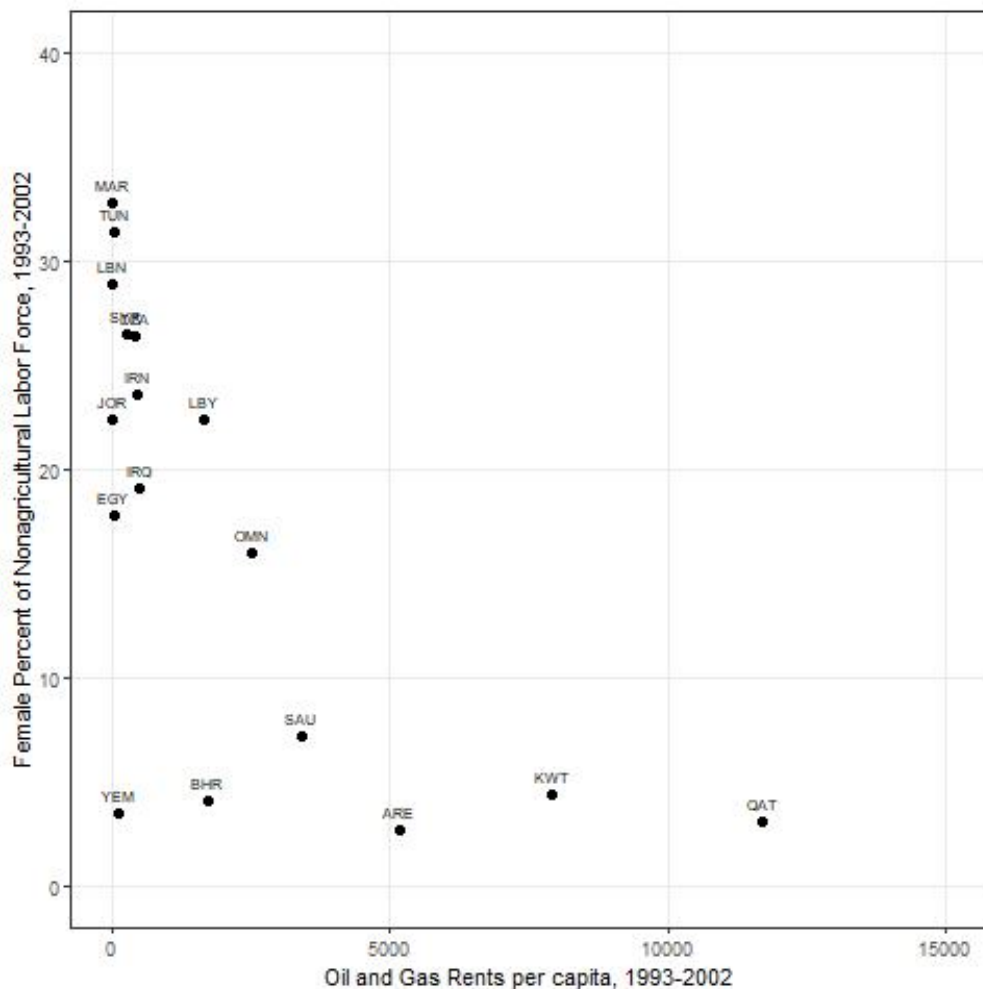


Figure 4

```

1 suffrage_years <- data.frame(id = c("DJI", "SYR", "LBN", "EGY", "TUN", "DZA",
  "IRN", "LBY", "MAR", "YEM", "BHR", "JOR", "IRQ", "OMN", "QAT", "SAU", "ARE",
  "KWT"), suffrage_year = c(1946, 1949, 1952, 1956, 1959, 1962, 1963,
  1964, 1967, 1970, 1973, 1974, 1980, 1994, 1999, 2005, 2005, 2005))
2
3 jpeg(file="figure4.jpeg")
4 suffrage_years %>% left_join(data_long %>% select(id, year, oil_gas_rentPOP),
  by = c("id" = "id", "suffrage_year" = "year")) %>%
5   mutate(oil_at_suffrage = ifelse(is.na(oil_gas_rentPOP), 0, oil_gas_rentPOP))
  %>%
6   ggplot(aes(x = oil_at_suffrage, y = suffrage_year, label = id)) +
7   geom_point() + geom_text(size = 3, vjust = -1) +
8   scale_x_continuous(name = "Oil and Gas Value per capita in Year of Female
  Suffrage (constant 2000 dollars)", limits = c(0, 20000), breaks = seq(0,
  20000, 5000)) +
9   scale_y_continuous(name = "Year of Female Suffrage", limits = c(1940, 2010),
  breaks = seq(1940, 2010, 10)) +
10  theme_bw() + theme(panel.grid.minor = element_blank())
11 dev.off()

```

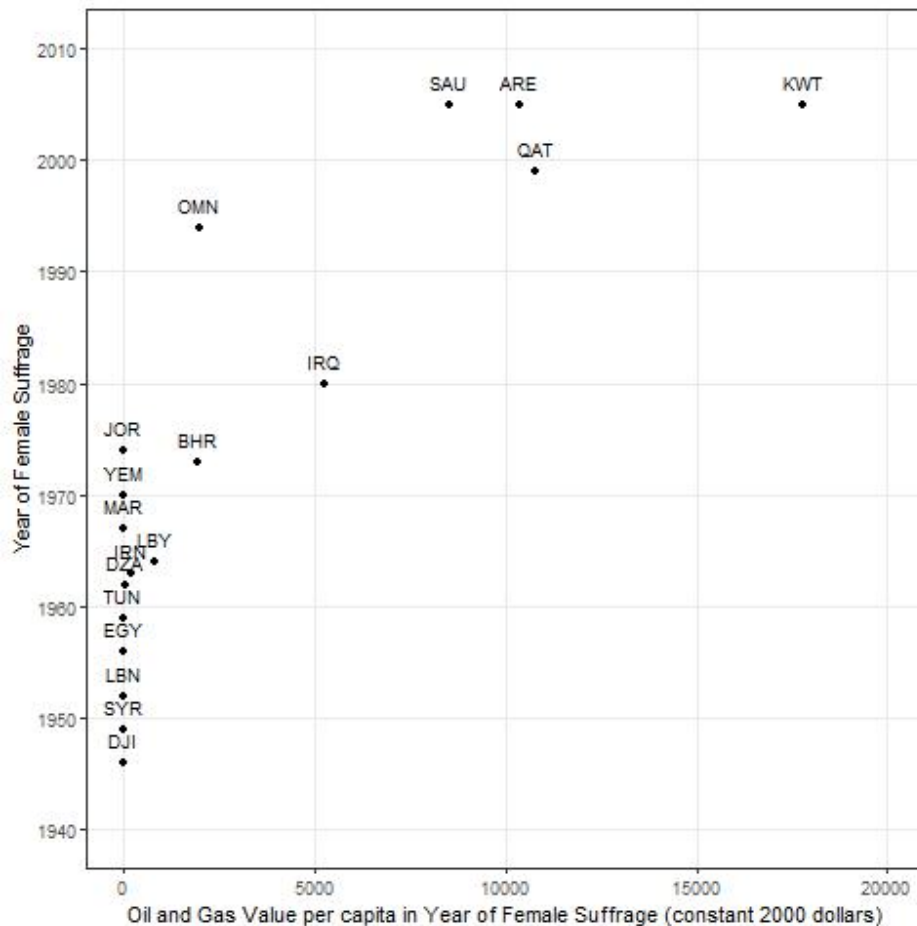


Figure 5

```

1 jpeg(file="figure5.jpeg")
2 data_short %>% filter(me_naf == 1) %>% select(id, female_seats) %>% left_join
  (oil_averages, by = "id") %>%
3   mutate(female_seats = ifelse(is.na(female_seats), 0, female_seats)) %>%
4   ggplot(aes(x = raw_oil, y = female_seats, label = id)) +
5   geom_point() + geom_text(size = 2.8, vjust = -1.1) +
6   scale_x_continuous(name = "Oil and Gas Rents per capita, 1993-2002", limits
7     = c(0, 15000), breaks = seq(0, 15000, 5000)) +
8   scale_y_continuous(name = "Parliamentary Seats Held by Women, 2003 (%)",
9     limits = c(0, 25), breaks = seq(0, 25, 5)) +
10  theme_bw() + theme(panel.grid.minor = element_blank())
11 dev.off()

```

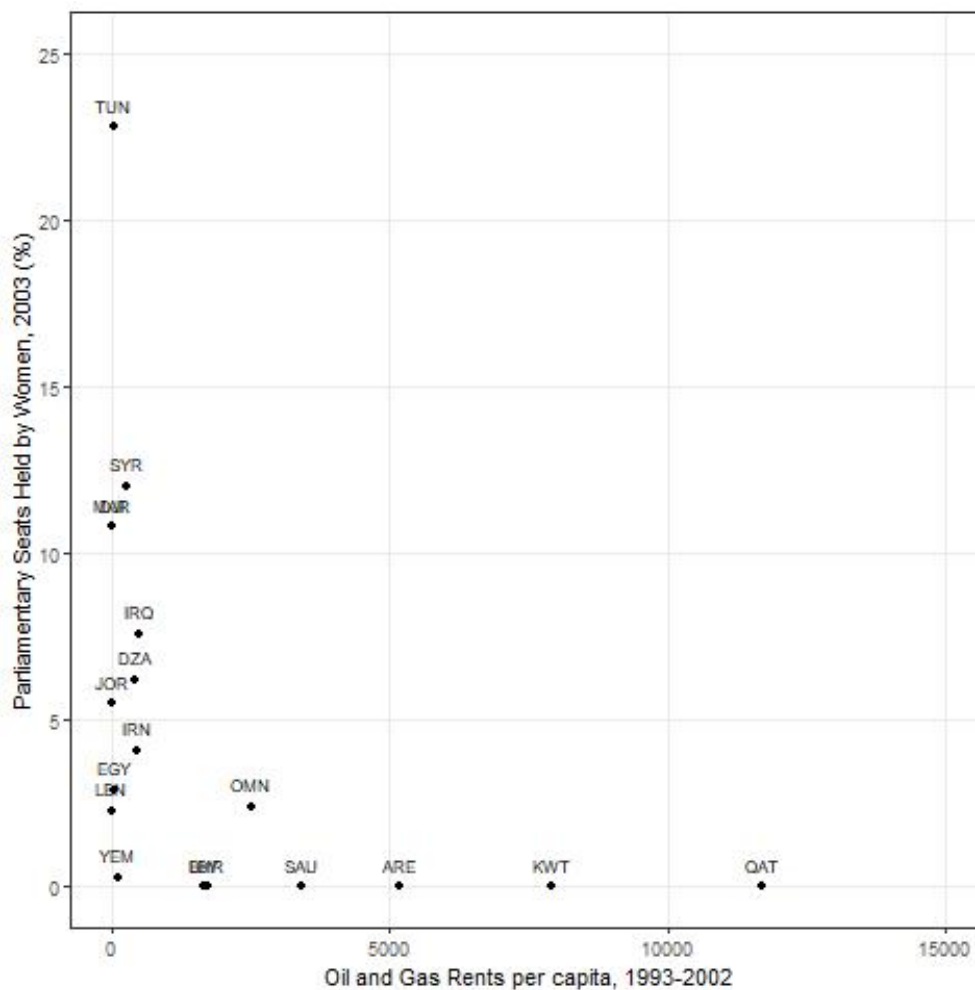
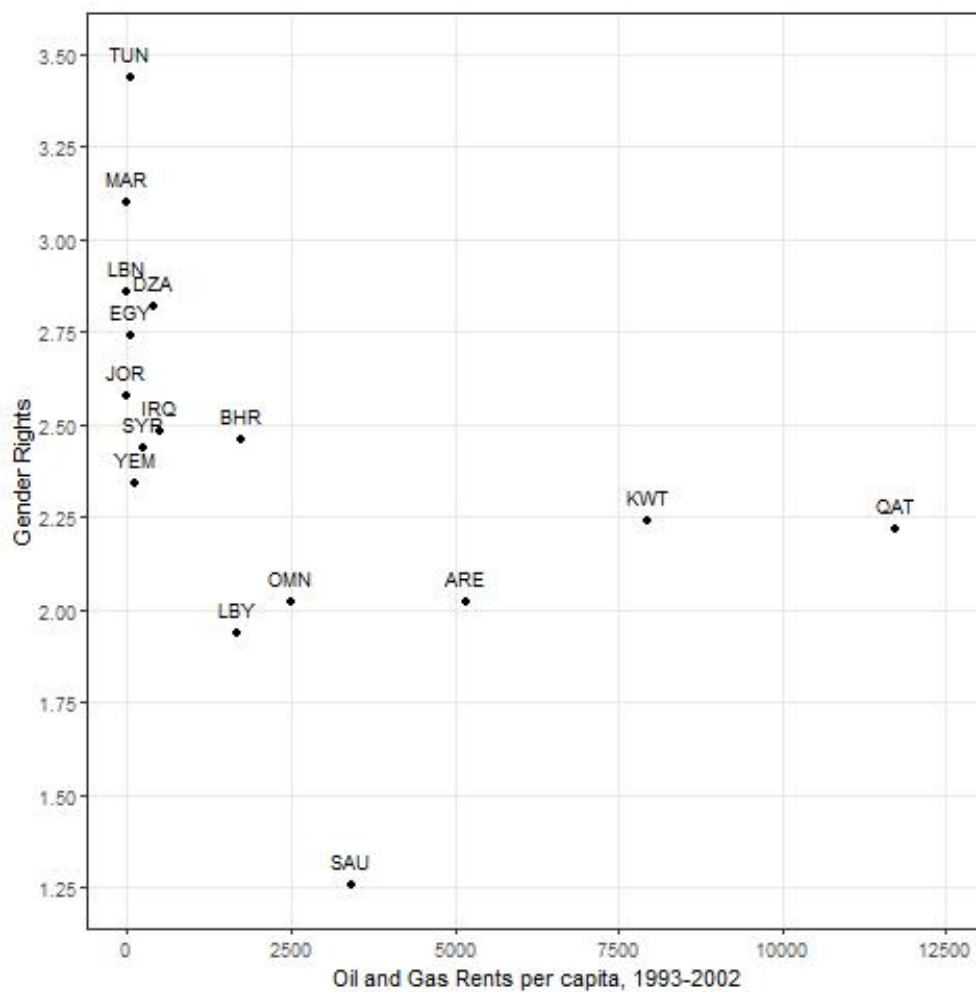


Figure 6

```

1 jpeg(file="figure6.jpeg")
2 data_short %>% filter(me_nafr == 1) %>% left_join(oil_averages, by = "id") %>%
3   ggplot(aes(x = raw_oil, y = gender_rights, label = id)) +
4   geom_point() + geom_text(size = 3, vjust = -1) +
5   scale_x_continuous(name = "Oil and Gas Rents per capita, 1993-2002", breaks
6     = seq(0, 12500, 2500)) +
7   scale_y_continuous(name = "Gender Rights", breaks = seq(1.25, 3.5, 0.25)) +
8   coord_cartesian(xlim = c(0, 12500), ylim = c(1.25, 3.5)) +
9   theme_bw() + theme(panel.grid.minor = element_blank())
10 dev.off()

```



Original Results: Interpretations

The tables show that oil rents per capita are significant in predicting female ministerial positions, parliamentary seats held by women, female nonagricultural labor force participation, and total female labor force participation. The plots visually suggest that (within the Middle East) higher oil and gas rents per capita is associated with lower gender rights, lower percentage of parliamentary seats held by women, and later year of female suffrage. The results support Ross's theory that oil rents are negatively associated with female labor force participation and political participation.

My Contribution

Looking at Natural Resources Rent

Ross theorizes that oil would negatively impact women's participation in the labor force based on the Dutch Disease Model. This theory is an extension of the resource curse, suggesting that an oil boom encourages an economic shift from trade to non-trade sector, which women are more likely to be excluded from. I questioned if this phenomena would apply to other natural resources, not just oil. To analyze this, I replaced oil rents with natural resource rents in the analysis.

Data Processing: For the natural resource rent data, if possible, the 2002 data would be used, corresponding with the original study. However, if the 2002 data is unavailable, the 1960-2002 (the years examined in the study) average will be used instead. The natural resource rent data was merged with Ross's data and the z-score was taken, to standardize the data in accordance with Ross's methodology.

```
1 # Load the Natural Resource Rent CSV
2 new_resource_data <- read_csv("natural_resource_rents.csv",
3                               skip = 4,
4                               show_col_types = FALSE)
5
6 # Years of the study (1960-2002)
7 study_years <- as.character(1960:2002)
8
9 # Prioritize 2002 data, but use the 1960-2002 average if there is no 2002 data
10 resource_clean <- new_resource_data %>%
11   select(id = 'Country Code', all_of(study_years)) %>%
12   # Convert year columns to numeric
13   mutate(across(all_of(study_years), as.numeric)) %>%
14   rowwise() %>%
15   # Calculate mean across the row for the period of the study
16   mutate(study_avg = mean(c_across(all_of(study_years)), na.rm = TRUE)) %>%
17   ungroup() %>%
```

```

18 # If 2002 is missing, use the average of all of the years in the study
19 mutate(rent_final = ifelse(!is.na('2002'), '2002', study_avg)) %>%
20 select(id, rent_final) %>%
21 filter(!is.na(rent_final))
22
23 # Merge the 'new_resource_data' with Ross's 'data_short' dataframe
24 data_short_updated <- data_short %>%
25 left_join(resource_clean, by = "id") %>%
26 # Use standardized data (Z-scores), to follow Ross's methodology
27 mutate(znat_res_rent = (rent_final - mean(rent_final, na.rm = TRUE)) /
28          sd(rent_final, na.rm = TRUE))

```

Table 1:

```

1 # Oil female labor model
2 female_labor_model_original <- lm(zlaborfemaleMOD ~ zlogGDPcap + zlogGDPcap2 +
3   zage15_64 +
4   zme_nafr + zcommie + zislam + zoil_gas -
5   rentPOP,
6   data = data_short_updated)
7 # Natural resource female labor model
8 female_labor_model_twist <- lm(zlaborfemaleMOD ~ zlogGDPcap + zlogGDPcap2 +
9   zage15_64 +
10   zme_nafr + zcommie + zislam + znat_res_rent,
11   data = data_short_updated)
12 stargazer(female_labor_model_original, female_labor_model_twist, type = "text"
13   ,
14   title="Female Labor Participation: Oil Model and Natural Resources
15   Model",
16   dep.var.labels = "Female Labor Force Participation",
17   covariate.labels = c("Income (log)",
18     "Income squared (log)",
19     "Working Age",
20     "Middle East",
21     "Communist",
22     "Islam",
23     "Oil Rents per Capita",
24     "Natural Resource Rents per Capita"))

```

	<i>Dependent variable:</i>	
	Female Labor Force Participation	
	(1)	(2)
Income (log)	−1.864*** (0.654)	−1.512** (0.678)
Income squared (log)	2.122*** (0.648)	1.765*** (0.669)
Working Age	−0.350*** (0.119)	−0.428*** (0.126)
Middle East	−0.326*** (0.090)	−0.378*** (0.093)
Communist	0.286*** (0.082)	0.307*** (0.086)
Islam	−0.139 (0.086)	−0.164* (0.089)
Oil Rents per Capita	−0.210*** (0.072)	
Natural Resource Rents per Capita		−0.070 (0.074)
Constant	−0.012 (0.060)	−0.013 (0.062)
Observations	167	163
R ²	0.427	0.402
Adjusted R ²	0.401	0.375
Residual Std. Error	0.776 (df = 159)	0.796 (df = 155)
F Statistic	16.898*** (df = 7; 159)	14.912*** (df = 7; 155)
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01

Table 2:

```
1 # Oil female seats model
2 female_seats_model_original <- lm(zfemale_seats ~ zlogGDPcap + zlogGDPcap2 +
   zage15_64 +
3       zme_nafr + zcommie + zislam + zoil_gas_rentPOP,
4       data = data_short_updated)
5
6 # Natural resources female seats model
7 female_seats_model_twist <- lm(zfemale_seats ~ zlogGDPcap + zlogGDPcap2 +
   zage15_64 +
8       zme_nafr + zcommie + zislam + znat_res_rent,
9       data = data_short_updated)
10
11 stargazer(female_seats_model_original, female_seats_model_twist,
12           type = "latex",
13           title="Female Seats (2002): Oil Model and Natural Resources Model",
14           dep.var.labels = "Parliamentary Seats Held by Women (percent), 2002"
15           ,
16           covariate.labels = c("Income (log)",
17                                "Income squared (log)",
18                                "Working Age",
19                                "Middle East",
20                                "Communist",
21                                "Islam",
22                                "Oil Rents per Capita",
23                                "Natural Resource Rents per Capita"))
```

	<i>Dependent variable:</i>	
	Parliamentary Seats Held by Women (percent), 2002	
	(1)	(2)
Income (log)	−1.748** (0.733)	−1.481* (0.765)
Income squared (log)	2.163*** (0.728)	1.894** (0.756)
Working Age	−0.090 (0.134)	−0.218 (0.144)
Middle East	−0.122 (0.100)	−0.160 (0.102)
Communist	0.140 (0.095)	0.194* (0.100)
Islam	−0.147 (0.100)	−0.167 (0.104)
Oil Rents per Capita	−0.259*** (0.079)	
Natural Resource Rents per Capita		−0.170** (0.085)
Constant	−0.015 (0.068)	−0.011 (0.071)
Observations	160	156
R ²	0.305	0.278
Adjusted R ²	0.273	0.244
Residual Std. Error	0.854 (df = 152)	0.879 (df = 148)
F Statistic	9.513*** (df = 7; 152)	8.128*** (df = 7; 148)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 3:

```

1 # Oil female minister model
2 female_minister_model_original <- lm(zfemale_min ~ zlogGDPcap + zlogGDPcap2 +
3   zage15_64 +
4   zme_nafr + zcommie + zislam + zoil_gas_
5   rentPOP,
6   data = data_short_updated)
7 # Natural resources female minister model
8 female_minister_model_twist <- lm(zfemale_min ~ zlogGDPcap + zlogGDPcap2 +
9   zage15_64 +
10   zme_nafr + zcommie + zislam + znat_res_rent,
11   data = data_short_updated)
12 stargazer(female_minister_model_original, female_minister_model_twist,
13   type = "latex",
14   title="Female Ministers (2002): Oil Model and Natural Resources
15   Model",
16   dep.var.labels = "Ministerial Seats Held by Women (%) 2002",
17   covariate.labels = c("Income (log)",
18     "Income squared (log)",
19     "Working Age",
20     "Middle East",
21     "Communist",
22     "Islam",
23     "Oil Rents per Capita",
24     "Natural Resource Rents per Capita"))

```

Female Ministers (2002): Oil Model and Natural Resources Model

	<i>Dependent variable:</i>	
	Ministerial Seats Held by Women (percent) 2002	
	(1)	(2)
Income (log)	−0.516 (0.810)	−0.446 (0.825)
Income squared (log)	0.894 (0.803)	0.817 (0.815)
Working Age	−0.151 (0.154)	−0.227 (0.158)
Middle East	−0.171 (0.118)	−0.179 (0.117)
Communist	−0.110 (0.107)	−0.070 (0.109)
Islam	−0.029 (0.120)	−0.036 (0.122)
Oil Rents per Capita	−0.145 (0.088)	
Natural Resource Rents per Capita		−0.145 (0.093)
Constant	−0.010 (0.076)	−0.005 (0.077)
Observations	154	151
R ²	0.174	0.165
Adjusted R ²	0.134	0.124
Residual Std. Error	0.933 (df = 146)	0.939 (df = 143)
F Statistic	4.383*** (df = 7; 146)	4.042*** (df = 7; 143)

Note:

*p<0.1; **p<0.05; ***p<0.01

Plots

Prepare data

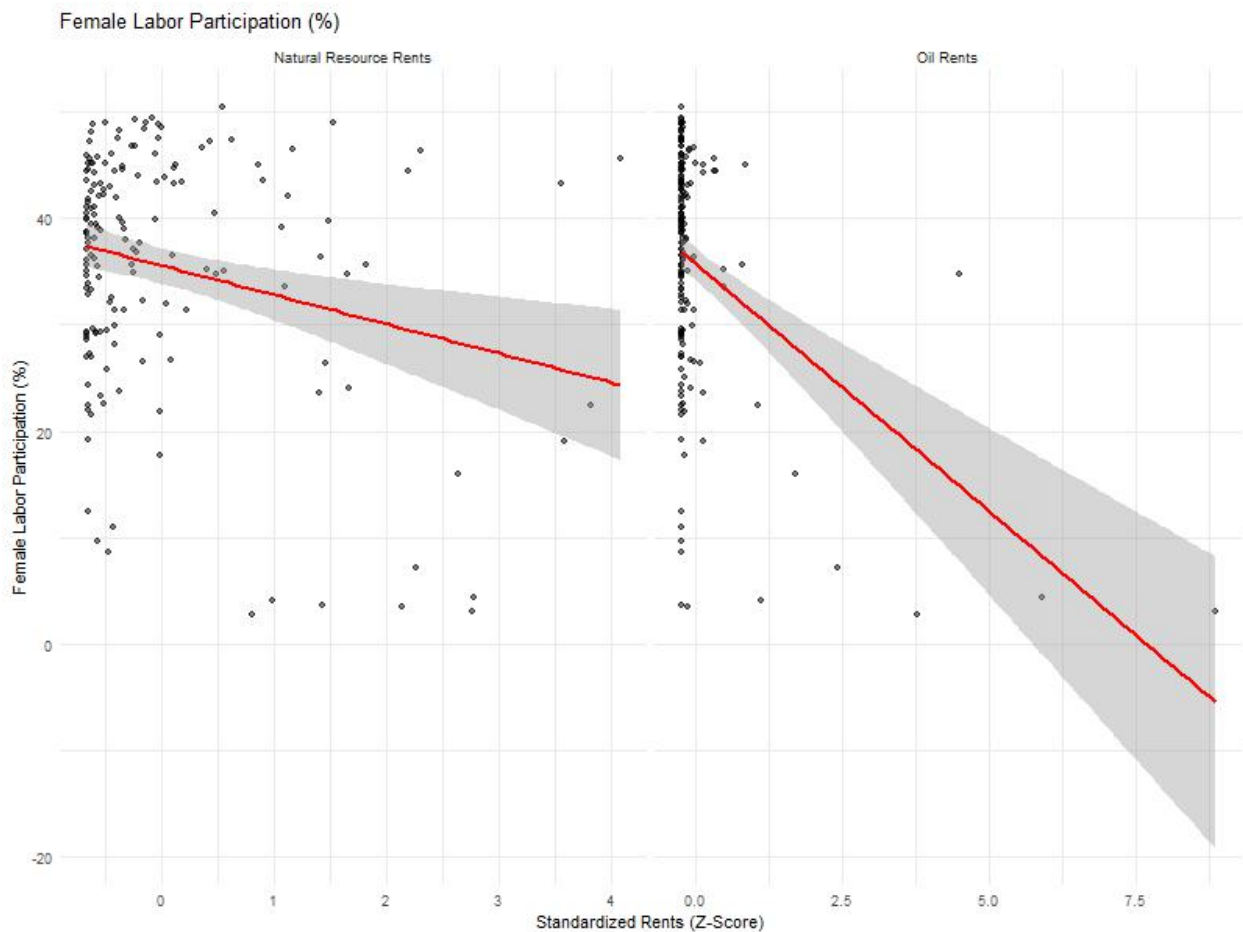
```
1 plot_data_tidy <- data_short_updated %>%
2   select(id, laborfemaleMOD, female_seats, female_min,
3     'Oil Rents' = zoil_gas_rentPOP,
4     'Natural Resource Rents' = znat_res_rent) %>%
5   pivot_longer(cols = c('Oil Rents', 'Natural Resource Rents'),
6     names_to = "Rent_Type",
7     values_to = "Z_Score")
```


Plot 1

```

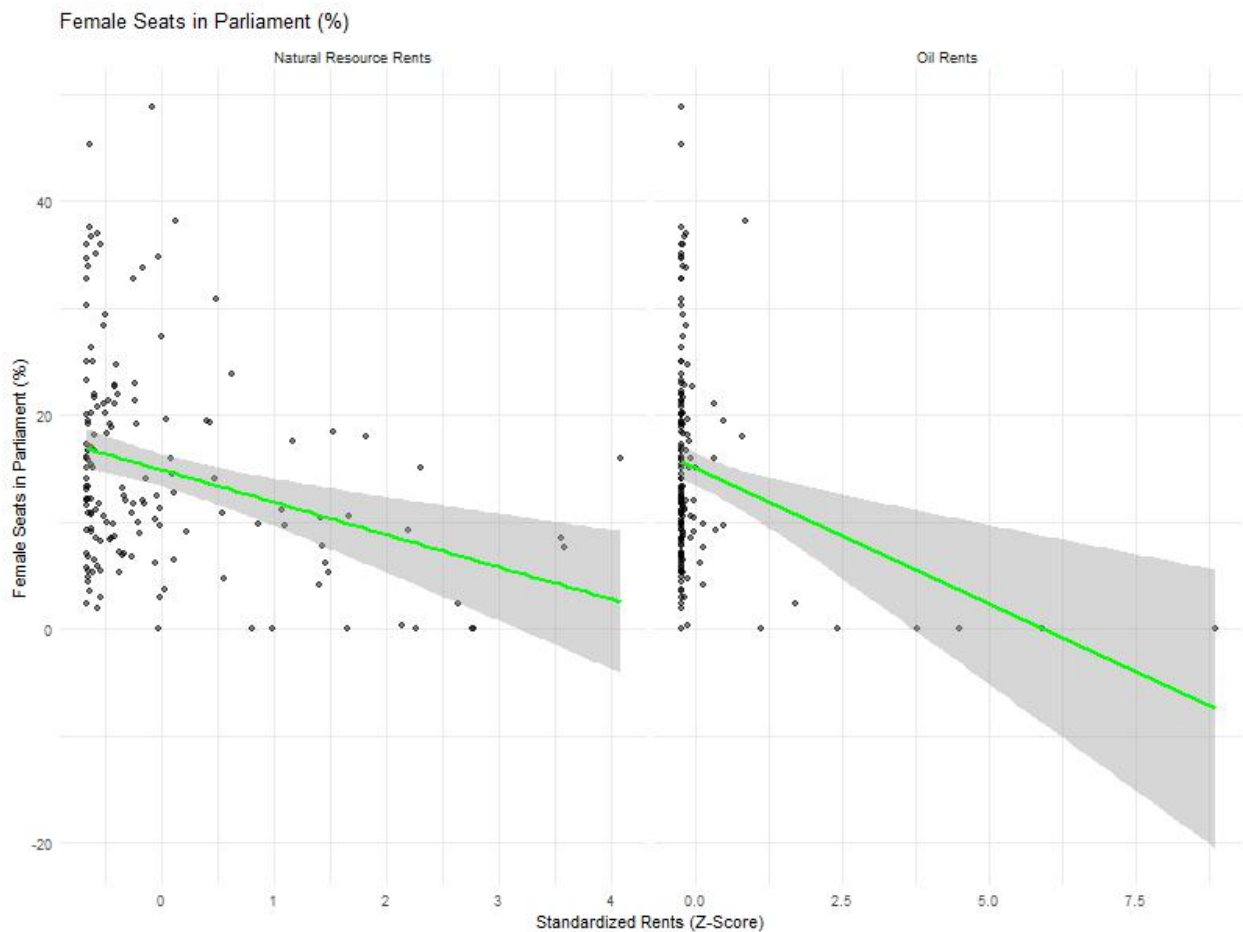
1 # 1. Female Labor Plot
2 jpeg("Female_Labor.jpeg", width=800, height=600)
3 ggplot(plot_data_tidy, aes(x = Z_Score, y = laborfemaleMOD)) +
4   geom_point(alpha = 0.5) +
5   geom_smooth(method = "lm", color = "red") +
6   facet_wrap(~Rent_Type, scales = "free_x") +
7   labs(title = "Female Labor Participation (%)",
8        x = "Standardized Rents (Z-Score)", y = "Female Labor Participation (%)")
9   theme_minimal()
10 dev.off()

```



Plot 2

```
1 # 2. Female Seats Plot
2 jpeg("Female_Seats.jpeg", width=800, height=600)
3 ggplot(plot_data_tidy, aes(x = Z_Score, y = female_seats)) +
4   geom_point(alpha = 0.5) +
5   geom_smooth(method = "lm", color = "green") +
6   facet_wrap(~Rent_Type, scales = "free_x") +
7   labs(title = "Female Seats in Parliament (%)",
8        x = "Standardized Rents (Z-Score)", y = "Female Seats in Parliament (%)")
9   theme_minimal()
10 dev.off()
```

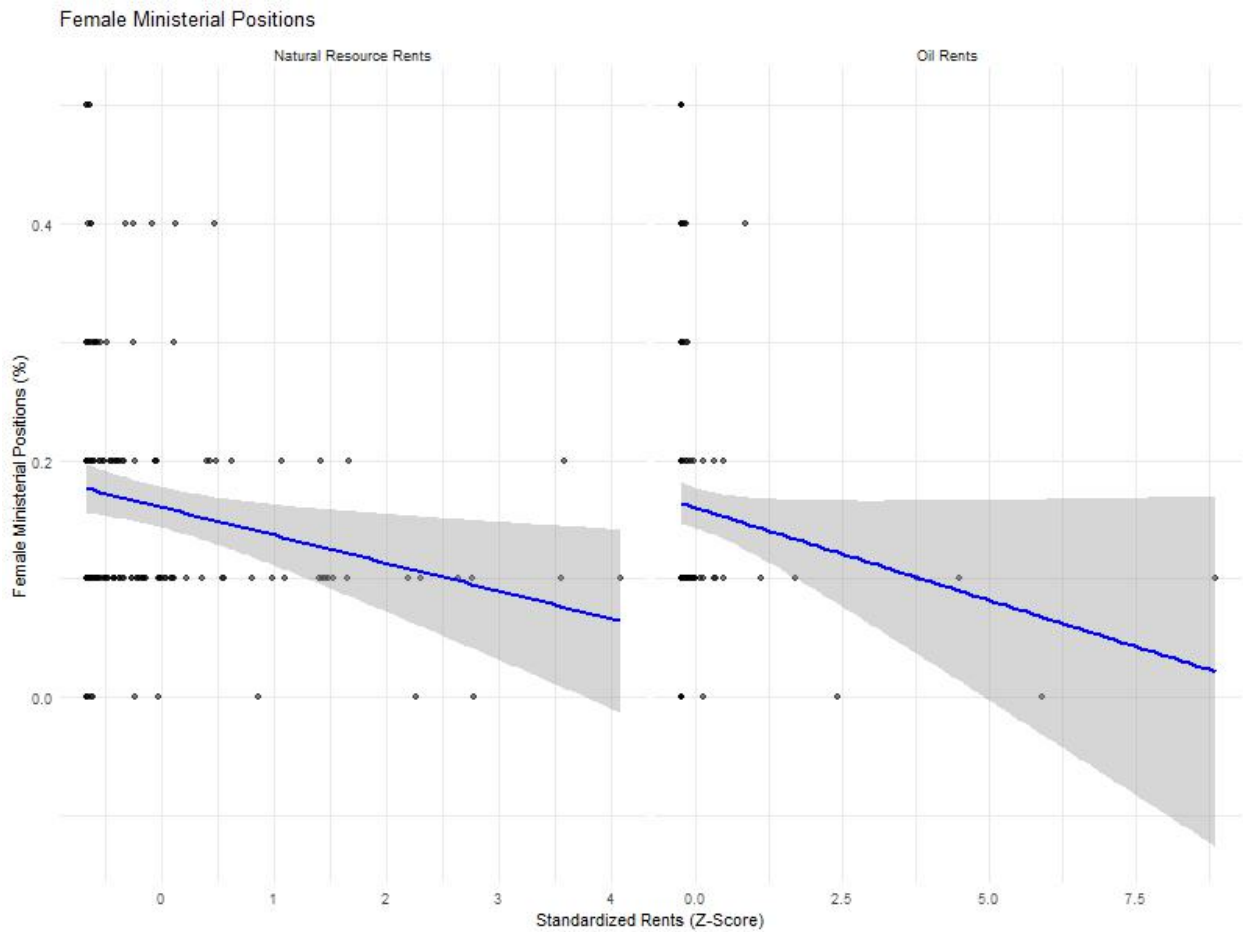


Plot 3

```

1 # 3. Female Ministers Plot
2 jpeg("Female_Ministers.jpeg", width=800, height=600)
3 ggplot(plot_data_tidy, aes(x = Z_Score, y = female_min)) +
4   geom_point(alpha = 0.5) +
5   geom_smooth(method = "lm", color = "blue") +
6   facet_wrap(~Rent_Type, scales = "free_x") +
7   labs(title = "Female Ministerial Positions",
8        x = "Standardized Rents (Z-Score)", y = "Female Ministerial Positions (%)") +
9   theme_minimal()
10 dev.off()

```



Conclusion

My results from Table 1 show that Natural Resource Rents per Capita are not significant in predicting Female Labor Force Participation. However, when Natural Resource Rents per Capita are included, Islam becomes significant at a 10 percent significance threshold (it was not significant in the oil rent model). The second table shows that both Oil Rents per Capita and Natural Resource Rents per Capita are significant in predicting the percentage of parliamentary seats held by women. However, Oil Rents per Capita are more significant. Additionally, in the model with Natural Resource Rents per Capita, Communist became significant at a 10 percent significance threshold. Finally, in Table 3, neither Oil Rents per Capita nor Natural Resource Rents per Capita are significant in predicting Ministerial Seats Held by Women.

The plot of Female Labor Participation shows that Oil Rents has a significantly steeper negative slope than Natural Resource Rents. However, Oil Rents has an only slightly steeper slope when plotting the percent of Female Seats in Parliament and an identical slope when plotting percent of Female Ministerial Positions.

While Natural Resource Rents, was not as significant as Oil Rents in predicting political and labor gender inequality, that result is informative. These results correspond with the assumption that Ross is making that the resource curse is not equal. Some natural resources (like oil) have a greater impact on female labor and political inclusion than others. However, further research is needed to examine the mechanisms behind this relationship and whether these results would hold in further studies. In particular, it would be informative to see if the relationship would be upheld if the study were updated with current data.